



Sukkur Institute of Business Administration University
Department of Computer Science

Software Construction & Development
BS – V (Software Engineering)
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LAB # 001

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Problem 001

Take 2 numbers from the console, your program must swap them without third variable and display on console.

Solution:

```
import java.util.Scanner;
class task01{
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter Number 01: ");
        int num1 = input.nextInt();
        System.out.print("Enter Number 02: ");
        int num2 = input.nextInt();

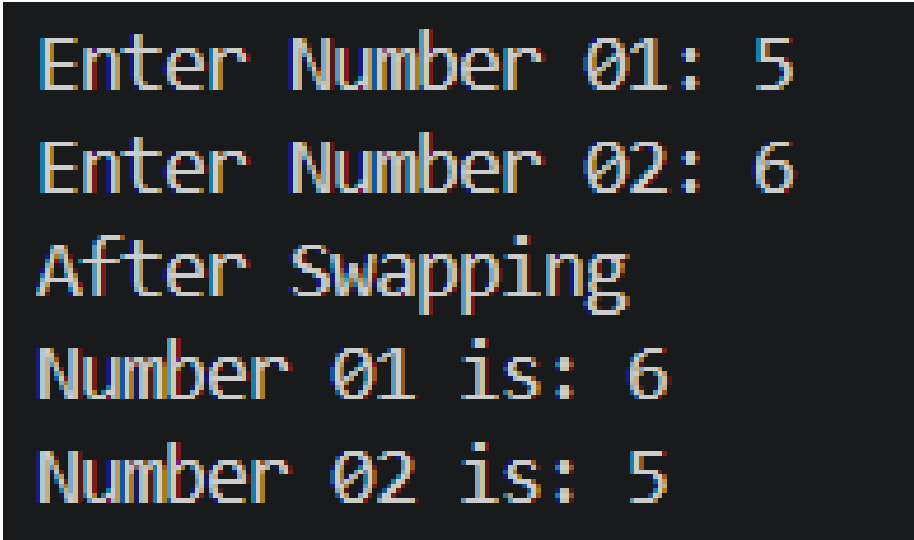
        // Swapping Logic
        num1 = num1 + num2;
        num2 = num1 - num2;
        num1 = num1 - num2;

        System.out.println("After Swapping");

        System.out.println("Number 01 is: " + num1);
        System.out.println("Number 02 is: " + num2);

        input.close();
    }
}
```

Output:

A screenshot of a terminal window with a black background and yellow text. The text shows the execution of the Java program: it prompts for two numbers, 5 and 6, then displays the result after swapping them, showing 6 for the first number and 5 for the second number.

```
Enter Number 01: 5
Enter Number 02: 6
After Swapping
Number 01 is: 6
Number 02 is: 5
```

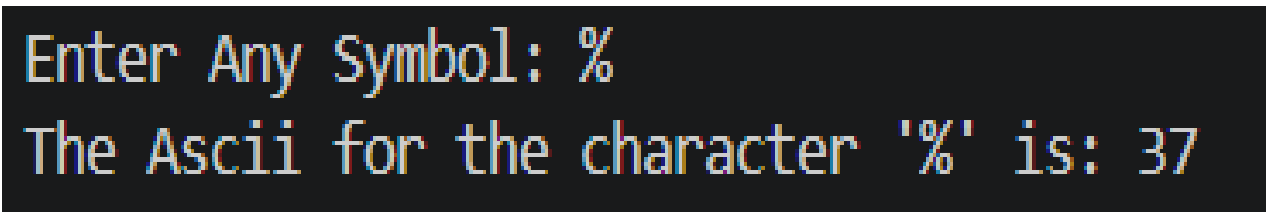
Problem 002

create a program in which user may enter any symbol, program must display the ASCII of that code.

Solution:

```
import java.util.Scanner;
public class task02 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter Any Symbol: ");
        char c = input.next().charAt(0);
        int symbol = (int)c;
        System.out.println("The Ascii for the character '"+c+"' is: " + symbol);
        input.close();
    }
}
```

Output:

A screenshot of a terminal window with a black background and light blue text. The first line shows the prompt "Enter Any Symbol: %" where the user has entered the percent symbol. The second line shows the output "The Ascii for the character '%' is: 37".

```
Enter Any Symbol: %
The Ascii for the character '%' is: 37
```

Problem 003

Use switch case to show user whether a user entered letter is a vowel or consonant or other symbol.

Solution:

```
import java.util.Scanner;
public class task03 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter Any Character to check if vowel or something else: ");
        char userInput = input.next().charAt(0);
        switch (Character.toLowerCase(userInput)) {
            case 'a','e','i','o','u':
                System.out.println(userInput + " IT IS A VOWEL!!!!");
                break;
            case 'w','y':
                System.out.println(userInput + " IT IS A SEMI VOWEL!!!!");
                break;
            default:
                if (Character.isLetter(userInput)) {
                    System.out.println(userInput + " IT IS A CONSONANT!!!!");
                } else {
                    System.out.println(userInput + " IT IS A SYMBOL!!");
                }
        }
        input.close();
    }
}
```

Output:

```
Enter Any Character to check if vowel or something else: b
b IT IS A CONSONANT!!!!
```

Problem 004

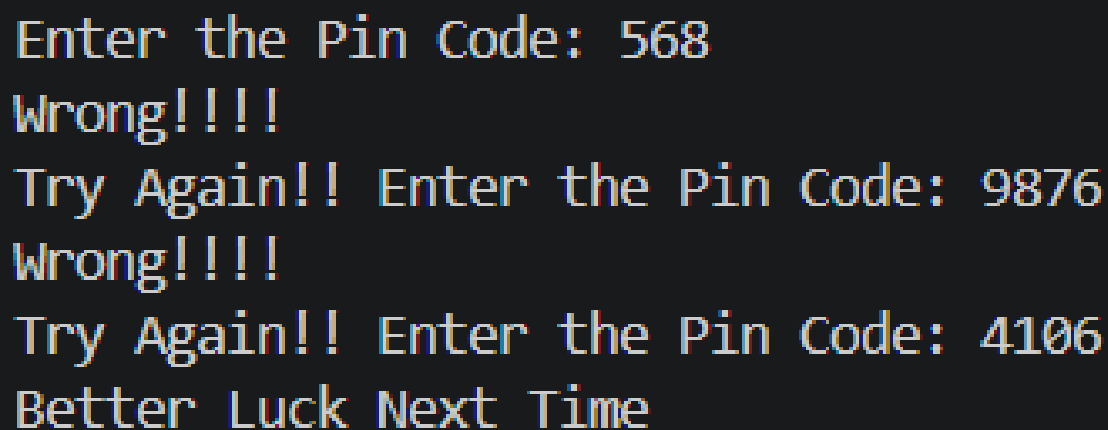
Create "PIN CODE ATM Verification Code". Use the do while loop and allow user to enter maximum 3 times for the pin code. If on the first attempt it is correct then show welcome screen. In rest of wrong pin code, allow other 2 times to insert pin code.

Solution:

```
import java.util.Scanner;

public class task04 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        final int pincode = 4106;
        int count = 0;
        do{
            count++;
            System.out.print("Enter the Pin Code: ");
            int userInput = input.nextInt();
            if(pincode==userInput){
                break;
            }else{
                System.out.println("Wrong!!!!");
                System.out.print("Try Again!! ");
            }
        } while(count<3);
        if (count==3) {
            System.out.println("Better Luck Next Time");
            return;
        }
        System.out.println("Welcome To My ATM");
        input.close();
    }
}
```

Output:



```
Enter the Pin Code: 568
Wrong!!!!
Try Again!! Enter the Pin Code: 9876
Wrong!!!!
Try Again!! Enter the Pin Code: 4106
Better Luck Next Time
```

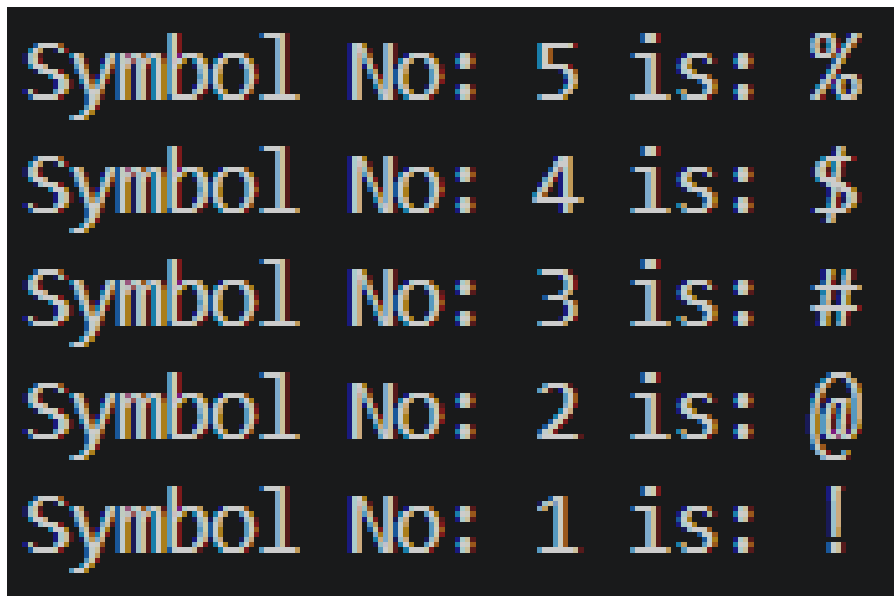
Problem 005

Create an array of 5 symbols, display it in reverse order.

Solution:

```
public class task05 {  
    public static void main(String[] args) {  
        char[] symbols = {'!', '@', '#', '$', '%'};  
        for (int i = (symbols.length-1); i >= 0 ; i--) {  
            System.out.println("Symbol No: " + (i+1) + " is: " + symbols[i]);  
        }  
    }  
}
```

Output:



The screenshot shows the output of the Java program, displaying five lines of text. Each line represents a symbol and its index in reverse order. The text is rendered in a yellow, monospaced font on a black background. The lines are: 'Symbol No: 5 is: %', 'Symbol No: 4 is: \$', 'Symbol No: 3 is: #', 'Symbol No: 2 is: @', and 'Symbol No: 1 is: !'.

```
Symbol No: 5 is: %  
Symbol No: 4 is: $  
Symbol No: 3 is: #  
Symbol No: 2 is: @  
Symbol No: 1 is: !
```

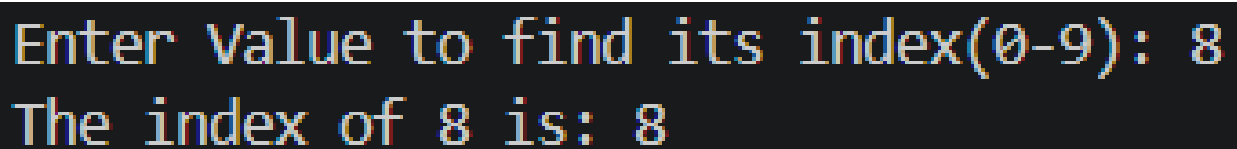
Problem 006

Find the value of a particular number from the array. Assume its size is 10.

Solution:

```
import java.util.Scanner;
public class task06 {
    public static void main(String[] args) {
        int arr[] = {0,1,2,3,4,5,6,7,8,9};
        Scanner input = new Scanner(System.in);
        System.out.print("Enter Value to find its index(0-9): ");
        int value = input.nextInt();
        int index=-1;
        for (int i = 0; i < arr.length; i++) {
            if(arr[i]==value){
                index=i;
            }
        }
        if(index== -1){
            System.out.println("The Entered Value Does Not Exists....");
        }else{
            System.out.println("The index of " + value + " is: " + index);
        }
        input.close();
    }
}
```

Output:

A screenshot of a terminal window with a black background and white text. The first line shows the prompt 'Enter Value to find its index(0-9):' followed by the user input '8'. The second line shows the output 'The index of 8 is: 8'.

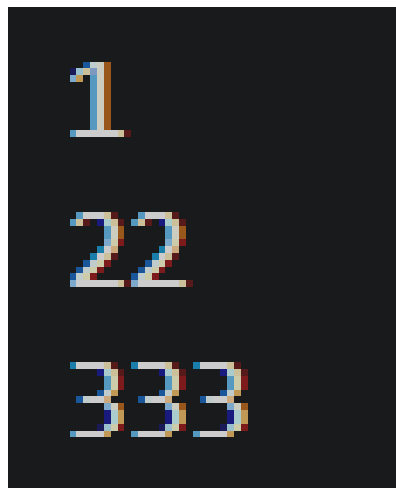
Problem 007

Create 2 d arrays, print the diagonal values using loops.

Solution:

```
public class task07 {  
    public static void main(String[] args) {  
        int[][] numbers = {{1,2,3},{11,22,33},{111,222,333}};  
        for(int i = 0; i<numbers.length ; i++){  
            System.out.println(numbers[i][i]);  
        }  
    }  
}
```

Output:



```
1  
22  
333
```

THE END